OHMWORK

TV · RADIO · REPAIR

- JOB PATH
- THE BENCH
- TICKETS
- TOOL WALL
- TOP TECHS
- PROFILE

JOB №0002 · UNIT 2

THE SILENT BOOMBOX

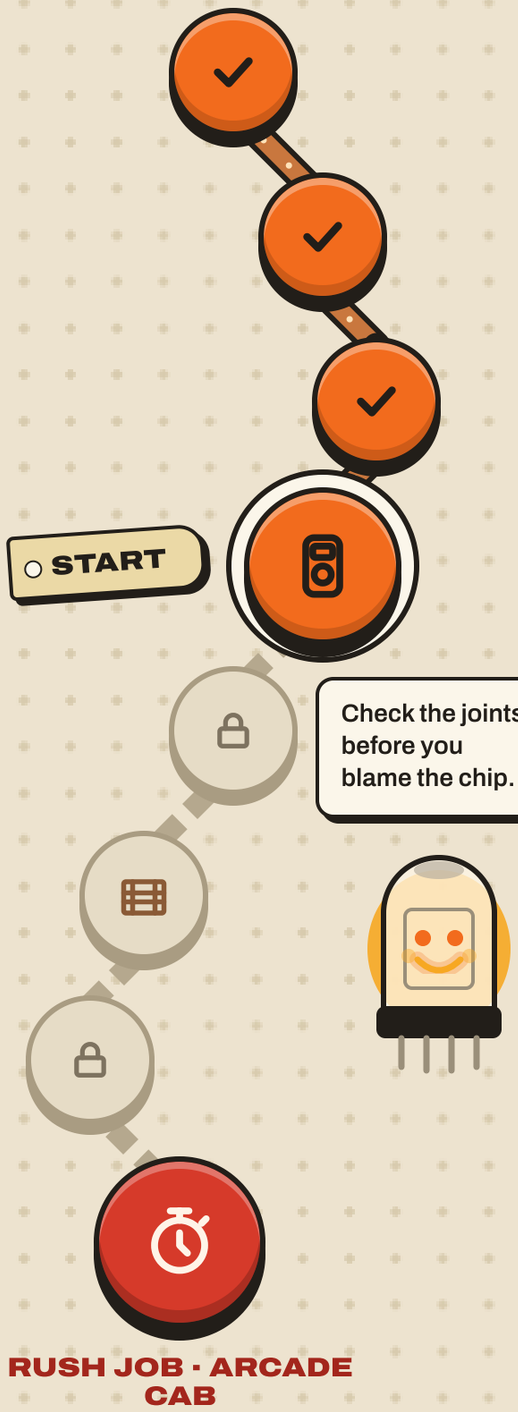
Series & parallel circuits — the left speaker went quiet.

SERVICE MANUAL



"Right side plays fine, left side's dead. My son's party is Saturday!" — Mrs. Delgado

STEPS 3/5



JOB №0003 · IN THE QUEUE

THE DETUNED SYNTH

Capacitors & RC timing — the band's arpeggiator drifts

TECH LV 3 | 12 | 5 | 340

TODAY'S TICKETS

NEW IN 06:12

№ T-118 · DAILY

Earn 40 XP

28/40

№ T-119 · DAILY

2 jobs, no blown fuses

1/2

№ T-120 · BENCH

Light an LED on the Bench

OPEN THE BENCH

№ T-117 · DAILY

~~Open the shop~~

DONE

TOOL WALL

TOOLS UNLOCK BENCH PARTS

- Multimeter — ON THE WALL
- Soldering iron — ON THE WALL
- Oscilloscope — JOB 04 · +PROBE
- Signal gen — JOB 06 · +AC SRC

BOSS · RUSH JOB

THE ARCADE CABINET

Lucky Lanes opens at 6. The cabinet won't boot. You've got 90 seconds on the clock.

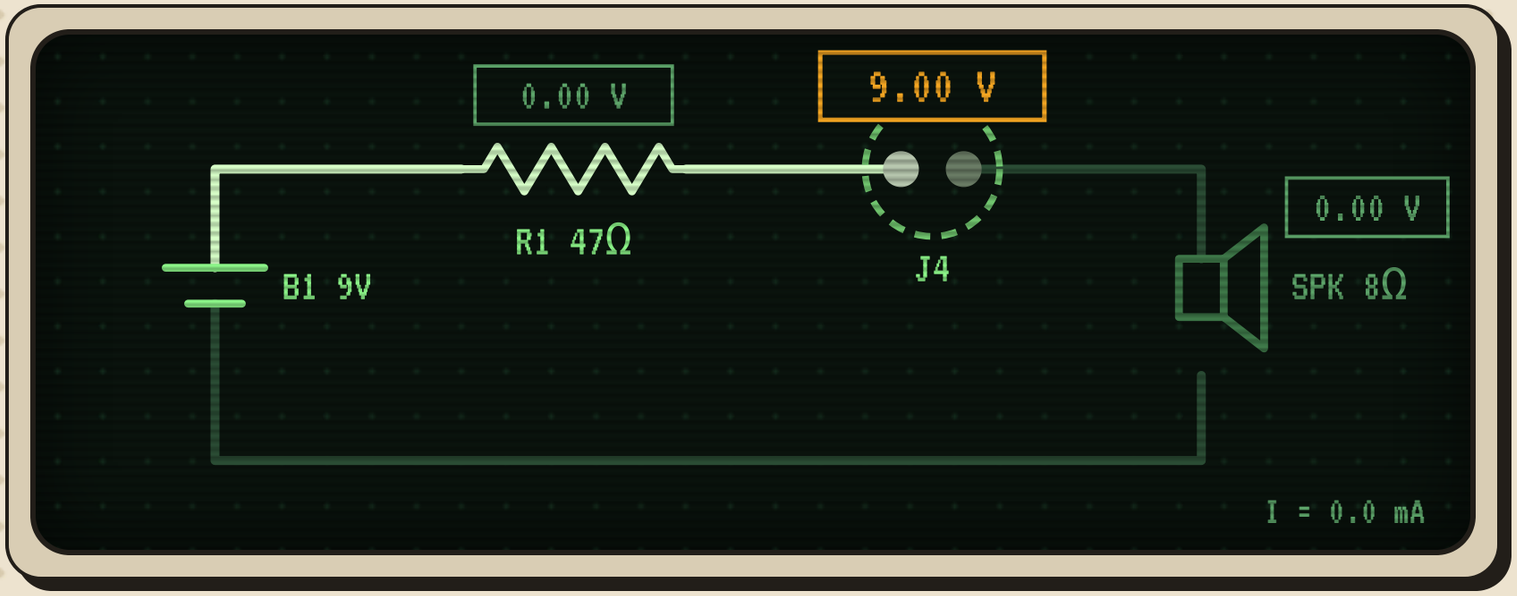
UNLOCKS IN 4 STEPS — X2 XP

JOB №0002 · STEP 4 · TROUBLESHOOT

The left speaker is dead silent. Where's the open circuit?

Your meter readings are on the screen. Tap the faulty part.

Where the volts pile up, the break is hiding.



1 R1

Joint J4

3 Speaker

4 Battery

NICE CATCH! +15 XP

With a break in a series loop, no current flows — so R1 and the speaker drop **0 V**, and the open joint gets the **whole 9 V** across it.

EXPLAIN

CONTINUE

THE BENCH · delgado-boombox.cir · SAVED

IMPORT FALSTAD .TXT | EXPORT | SHARE BUILD

PAUSE | RUN t=0.150s | SIM SPEED | DOT SPEED | VOLTS | CURRENT | FAULTS ON

PARTS BINS DRAG → SCREEN

Part no. or name…

POWER ×4

- B Battery
- V DC source
- SIGNAL GEN · JOB 06
- G Ground

PASSIVE ×4

- R Resistor
- C Capacitor
- L Inductor
- P Pot

SEMICONDUCTOR ×4

- D Diode
- IN HAND
- Q NPN
- M MOSFET

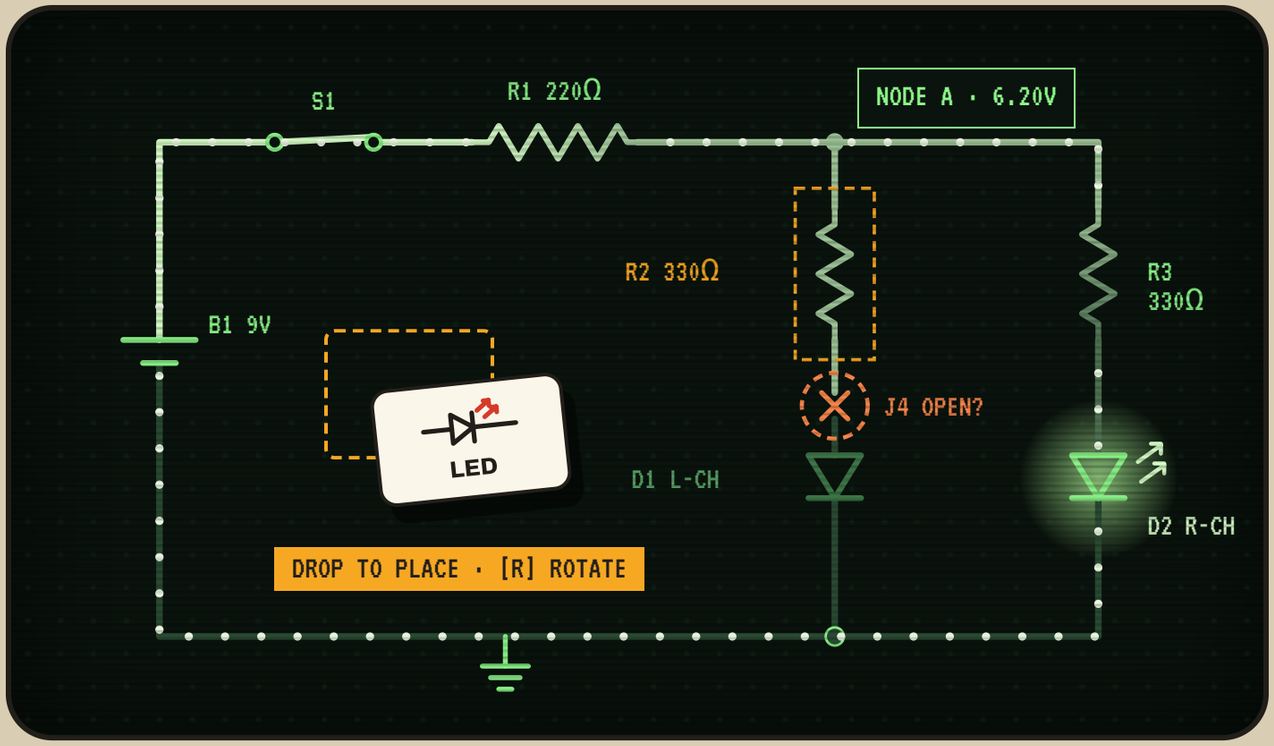
LOGIC GATES ×4

SWITCHES ×2

METERS & OUT ×4

OHMWORK BENCH MONITOR · MODEL 84

− 100% +



0V 9V · CURRENT · SUSPECTED FAULT

PROBES LEFT vs RIGHT

D1 · LEFT	D2 · RIGHT
0.00 V	2.00 V

SCOPE TRACE

Hang the oscilloscope on the tool wall (Job 04) to see waveforms.

INSPECTOR

R2 · RESISTOR

NO CURRENT · 0.0 mA

330 Ω

ORANGE · ORANGE · BROWN · GOLD

Resistance

330 Ω

FAULT FINDER

Continuity **OPEN** between R2 and D1 at joint **J4**. Looks like a cold solder joint.

REFLOW J4

ROTATE | FLIP

LESSON

Finding an open circuit →

REPAIR SHOP '84 · KIT

A 1980s TV & radio bench. Ink-outlined keycaps with hard shadows, test-gear cream, green-phosphor screens and manila repair tickets.

COLORS

SIGNAL	AMBER	BROWN	COPPER	INK
#F26B1D	#F7A823	#7A4A2A	#C9773E	#221E19

PANEL	MANILA	PHOSPHOR	LIVE	FAULT
#FBF6EA	#EBD9A6	#8DF78A	#2F9E44	#D63A2A

TYPE

SILENT BOOMBOX

Archivo Expanded 900 · headings, keys

An open joint gets the whole supply across it.

Archivo 400 · body

JOB №0002 · T-118

Space Mono · tickets, labels

9.00 V

VT323 · screens

MASCOT



TUBS, THE TUBE

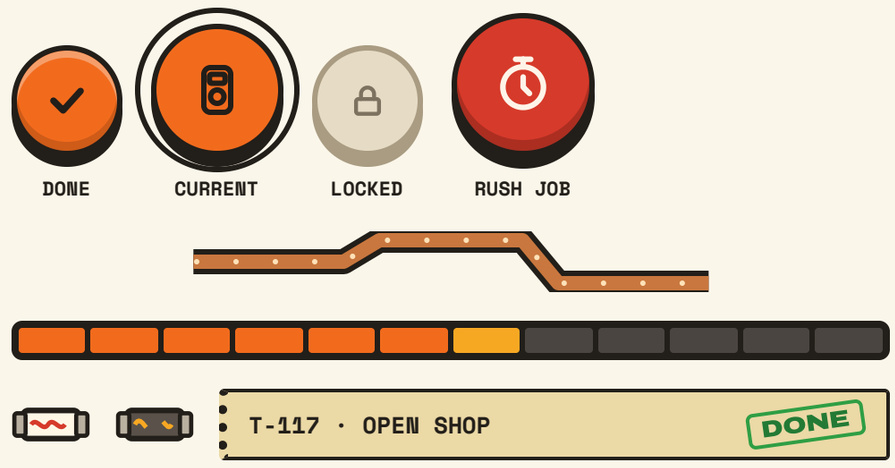
Warm, old-school, a little dramatic about cold joints.

KEYCAPS · 2.5 px ink outline, 5 px ledge; pressed sinks onto it

START JOB · PRESSED · CHECK · SERVICE MANUAL

CONTINUE · GOT IT · EXPLAIN · 2 Joint J4

PATH & HUD



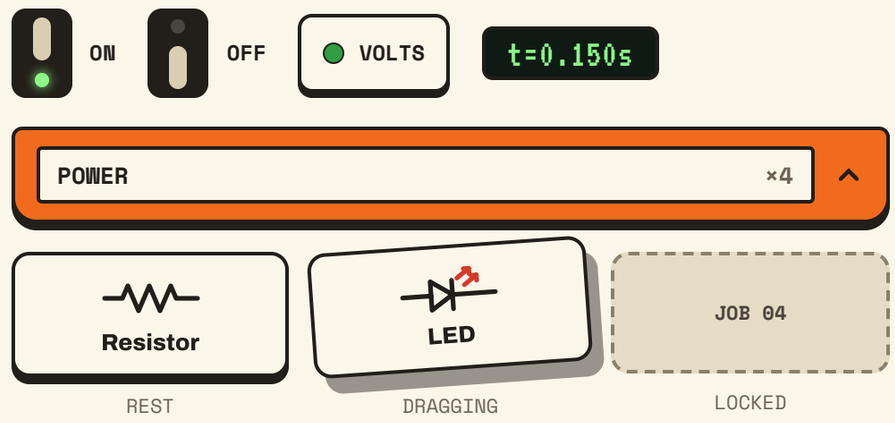
ANSWERS & FEEDBACK

R1 (DEFAULT) · Joint J4 (SELECTED) · ✓ Joint J4 (CORRECT) · ✕ Speaker (WRONG)

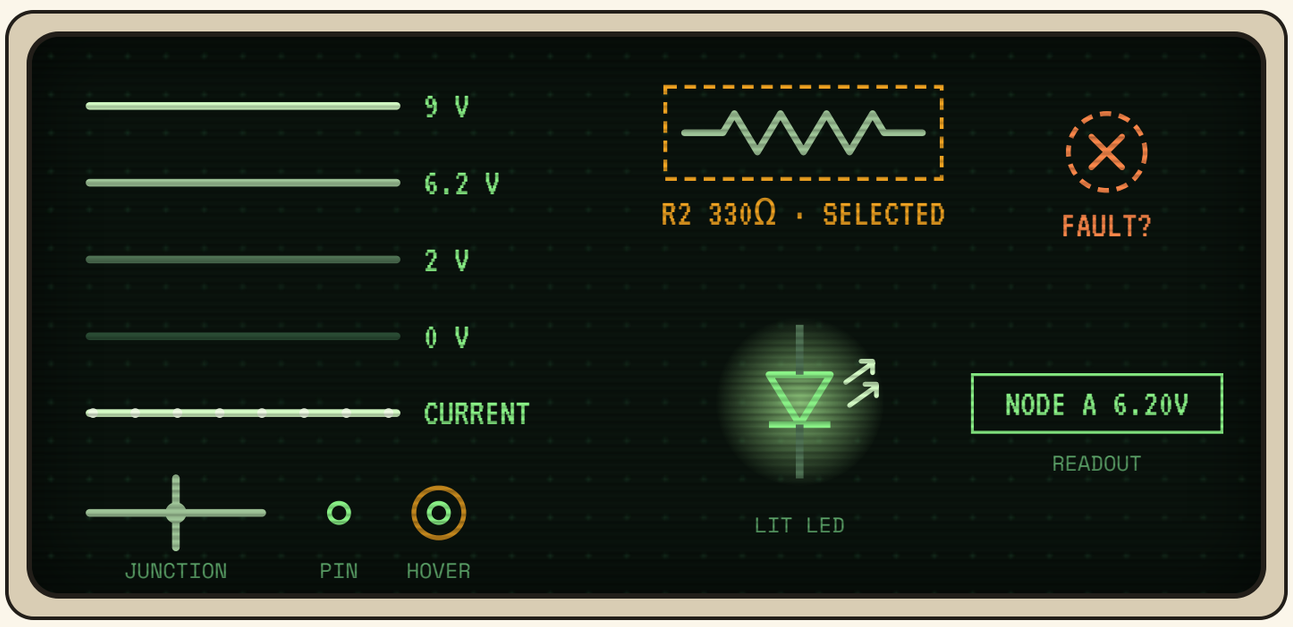
NICE CATCH! +15 XP — CONTINUE

FUSE BLOWN — ANSWER: JOINT J4 — GOT IT

CONTROLS & BINS



CIRCUIT ON THE CRT · Falstad's logic, monochrome phosphor: brighter = higher voltage



PARTS LIBRARY

Battery · DC source · AC source · Ground · Resistor · Pot · Capacitor · Electrolytic · Inductor

Diode · LED · NPN · MOSFET · Toggle · Push button · Solder joint · Speaker · Voltmeter